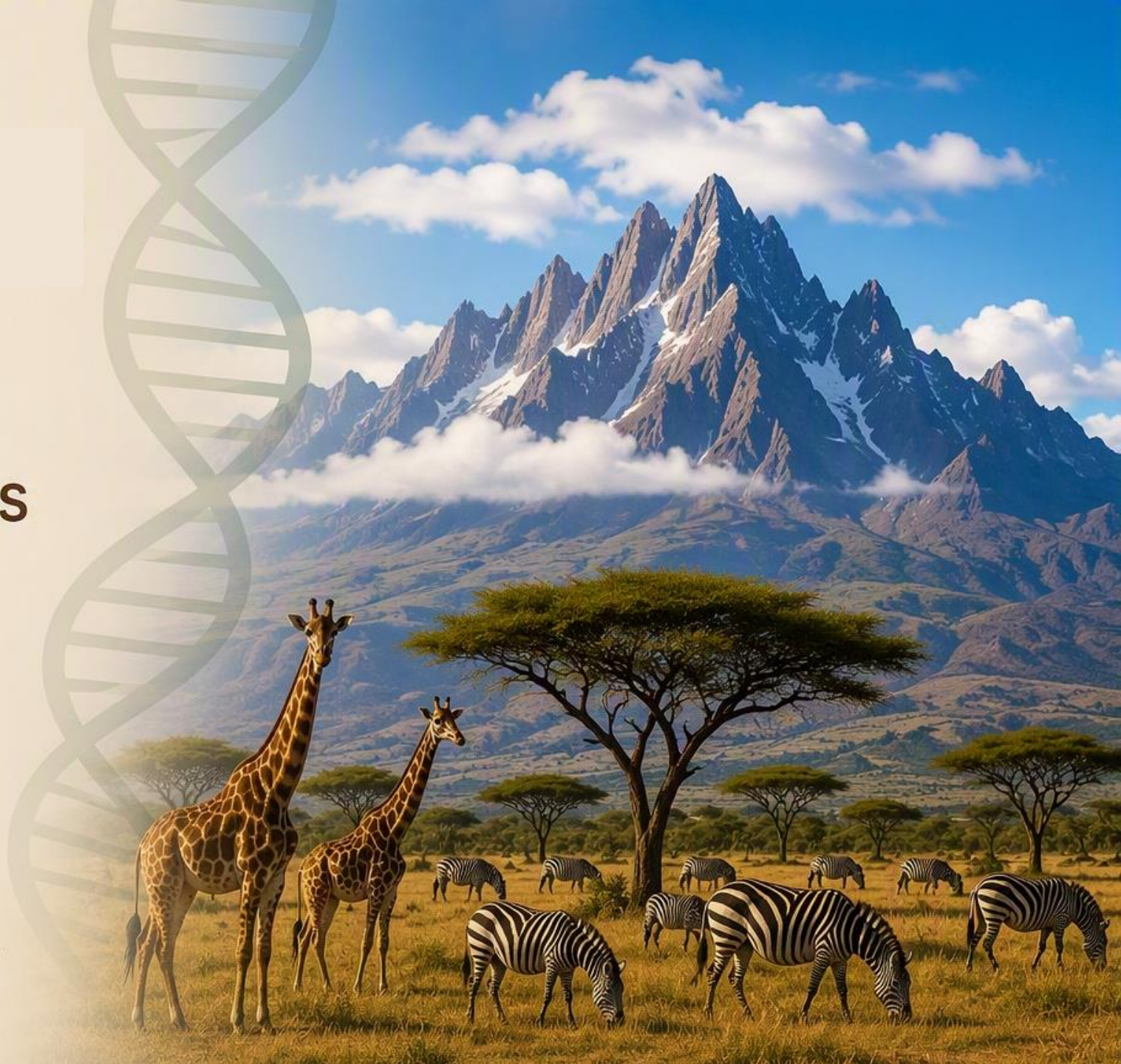
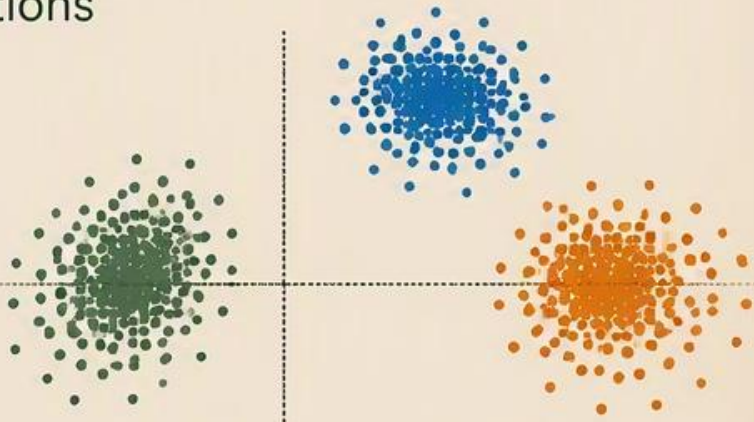


$$F_{ST}$$

IN POPULATION GENETICS

Understanding genetic similarity
within and differentiation between
populations



Today's class

Both methods compare observed allele sharing to expected sharing under a demographic model.

Relatedness

How related are two individuals?
IBD sharing vs expectation under
random mating

Relatedness asks: how much more do two individuals share than expected for random individuals?

FST

How differentiated are
populations?
Allele frequency variance among
groups

FST asks: how much do allele frequencies differ among populations compared to within?

FST: Variance in Allele Frequencies

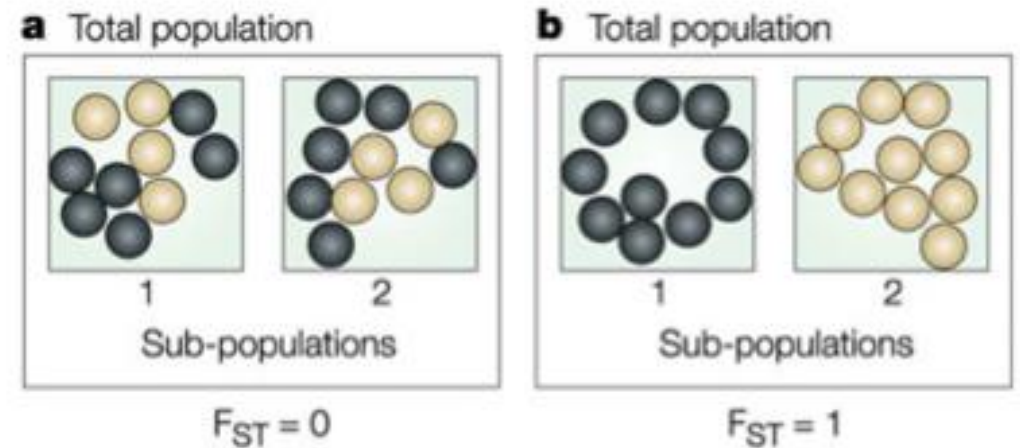
FST – fixation index

How differentiated are
populations?
Allele frequency variance among
groups

**FST measures the proportion of genetic variance
found among populations rather than within them**

Low Fst – No differentiation between the two
populations in this allele

High Fst – Two populations are fixed for different alleles
– Maximum differentiation



FST - Genetic differentiation

Interpretation

FST ≈ 0

Subpopulations are genetically identical

FST ≈ 0.05

Low diff. (e.g., European countries)

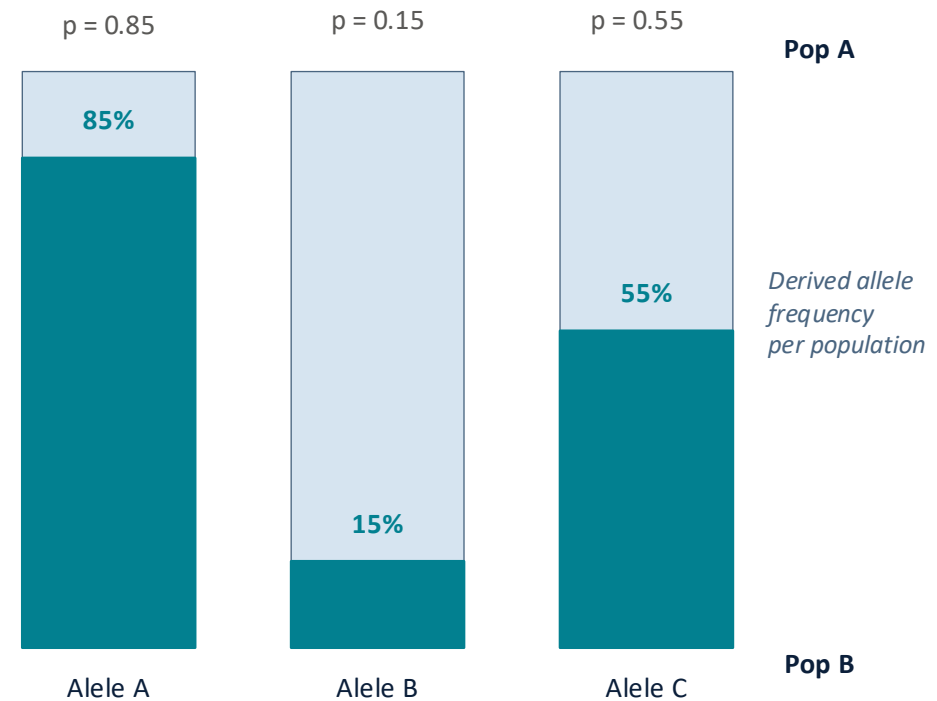
FST ≈ 0.15

Moderate (e.g., human continental groups)

FST ≈ 1

Complete fixation – different alleles fixed

FST $\sim$ proportion of variance between groups



High FST between Pop A & Pop B: allele nearly fixed in A, rare in B.

Weir-Cockerham vs Hudson Estimators

Weir-Cockerham (1984)

Method-of-moments estimator

- Accounts for sample size and number of populations
- Decomposes variance into **within-pop**, **among-pop**, and **total components**
- Handles unequal sample sizes (with caveats)
- Traditional standard for pairwise and global FST
- Implemented in VCFtools, hierfstat, PLINK

Wright (1943) / Weir & Cockerham (1984)

$$F_{ST} = (H_T - H_S) / H_T$$

H_T = expected heterozygosity in total population

H_S = mean expected heterozygosity within subpopulations

Hudson (1992)

Ratio of numerator/denominator

- Designed for **pairwise population comparison**
- More stable for **small sample sizes**
- Used in population genetics software (e.g., EIGENSOFT)
- Often preferred for **per-site FST** in selection scans
- Less biased for pairwise comparisons with unequal n

Hudson (1992)

$$F_{ST} = 1 - H_w / H_b$$

H_w = mean differences within-population

H_b = mean differences between-population

What causes FST?

What causes FST?

● = allele A

● = allele a

1 MUTATION

New alleles arise.



- Changes in DNA create new alleles.
- Source of all new genetic variation.
- Usually occurs rarely.

2 GENE FLOW

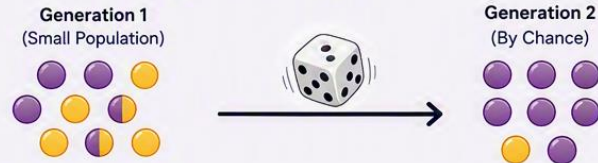
Alleles move between populations.



- Movement of individuals (and their alleles) between populations.
- Introduces new alleles or increases existing ones.
- Increases genetic variation within populations.

3 GENETIC DRIFT

Random changes in allele frequencies.



- Chance events cause alleles to increase or be lost.
- Strongest effect in small populations.
- Can lead to loss of genetic variation.

4 NATURAL SELECTION

Differential survival and reproduction.



- Individuals with advantageous alleles survive and reproduce more.
- Beneficial alleles increase; harmful alleles decrease.
- Leads to adaptation to the environment.



TOGETHER, THESE FORCES SHAPE THE GENETIC MAKEUP OF POPULATIONS OVER TIME.

Mutation creates variation | Gene flow moves variation | Genetic drift shuffles variation | Natural selection favors variation

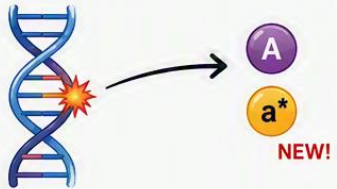
What causes FST?

● = allele A

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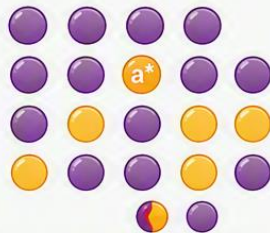
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Population



2 GENE FLOW

Alleles move between populations.



- Movement of individuals (and their alleles) between populations.
- Introduces new alleles or increases existing ones.
- Increases genetic variation within populations.

3 GENETIC DRIFT

Random changes in allele frequencies.

Generation 1
(Small Population)



Generation 2
(By Chance)



- Chance events cause alleles to increase or be lost.
- Strongest effect in small populations.
- Can lead to loss of genetic variation.

4 NATURAL SELECTION

Differential survival and reproduction.



- Individuals with advantageous alleles survive and reproduce more.
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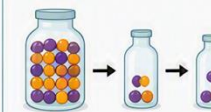
TOGETHER, THESE FORCES SHAPE THE GENETIC MAKEUP OF POPULATIONS OVER TIME.

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5 DEMOGRAPHIC HISTORY AND POPULATION STRUCTURE

Changes in population size or structure affect allele frequencies.

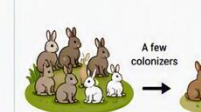
Population Bottlenecks



A drastic reduction in population size by chance reduces genetic variation. When the population grows again, allele frequencies may be very different from the original population.

- ✓ Key idea: Chance events during bottlenecks can strongly change allele frequencies.

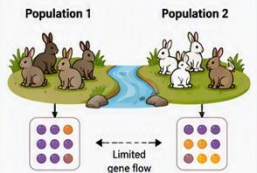
Founder Effects



A new population is started by a small number of individuals. Their allele frequencies may differ from the original population, especially by chance.

- ✓ Key idea: Chance sampling of a few individuals leads to different allele frequencies.

Population Structure
(Subdivision & Limited Gene Flow)



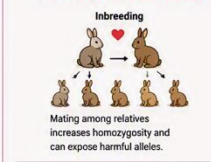
When populations are separated and have limited migration, they accumulate genetic differences over time.

- ✓ Key idea: Reduced gene flow allows populations to evolve differently.

6 NON-RANDOM MATING AND SOCIAL STRUCTURE

Mating patterns and social organization affect allele frequencies.

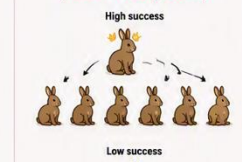
Non-Random Mating
(Inbreeding / Assortative Mating)



Similar mates Dissimilar mates
Individuals choose mates with similar (or dissimilar) traits, changing the distribution of alleles.

- ✓ Key idea: Who mates with whom affects genotype frequencies.

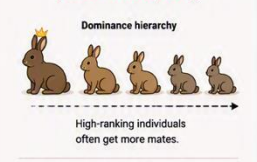
Unequal Reproductive Success
(Variance in Mating Success)



When some individuals leave many more offspring than others, their alleles become more common in the next generation.

- ✓ Key idea: Differences in reproductive success change allele frequencies.

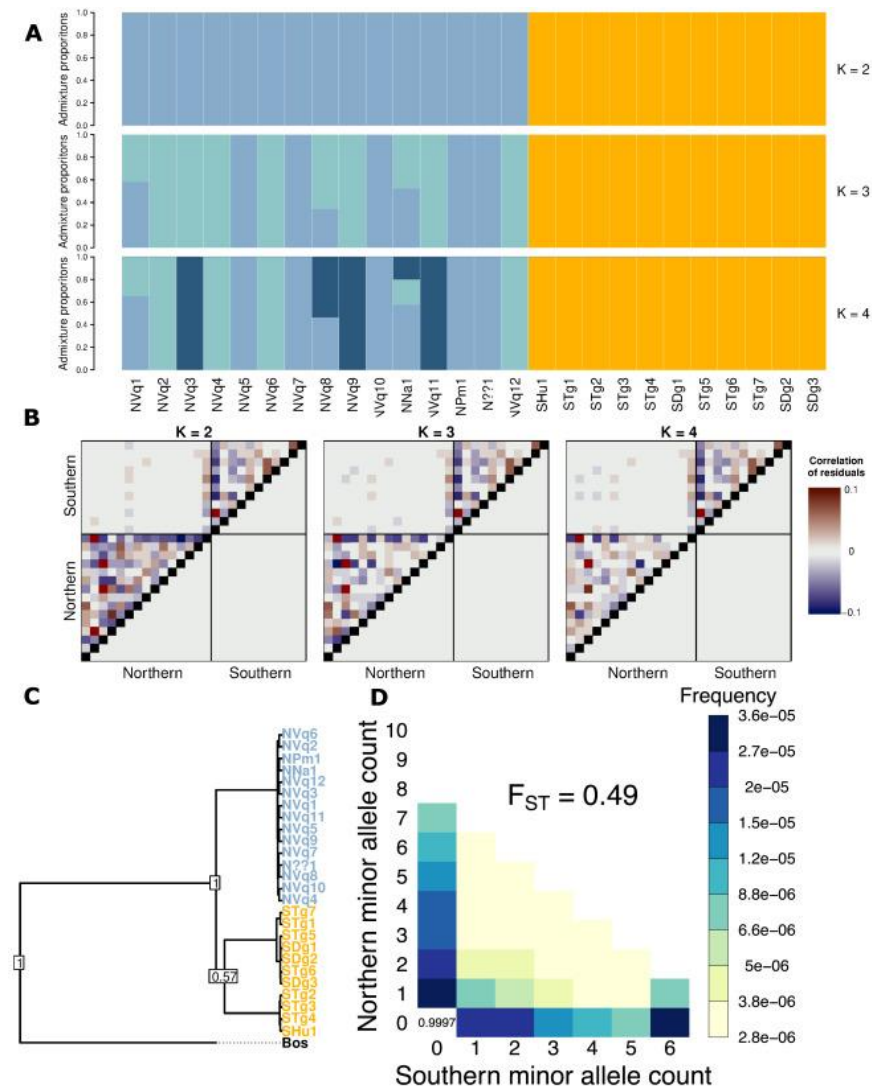
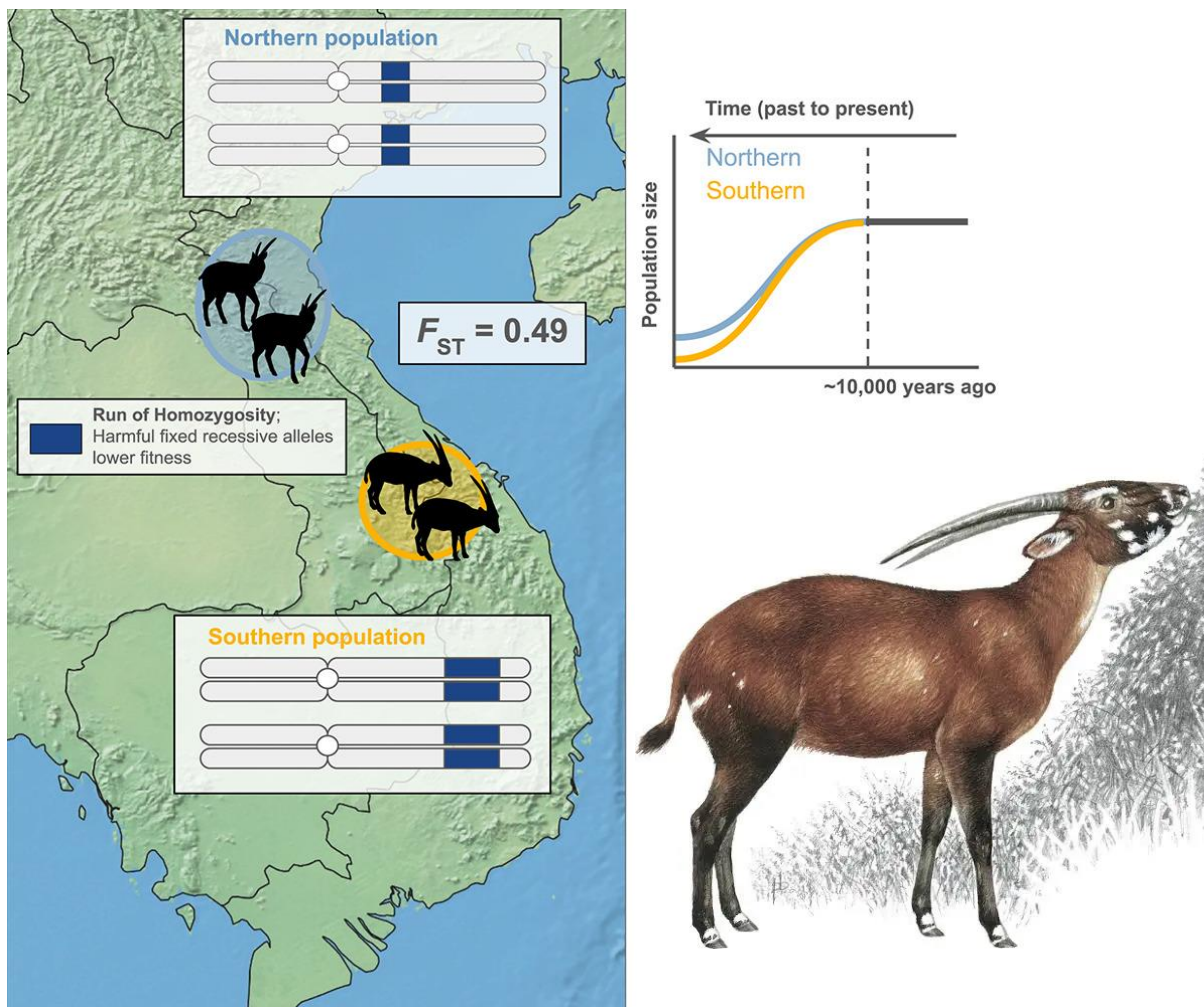
Social Structure
(Hierarchies & Group Living)



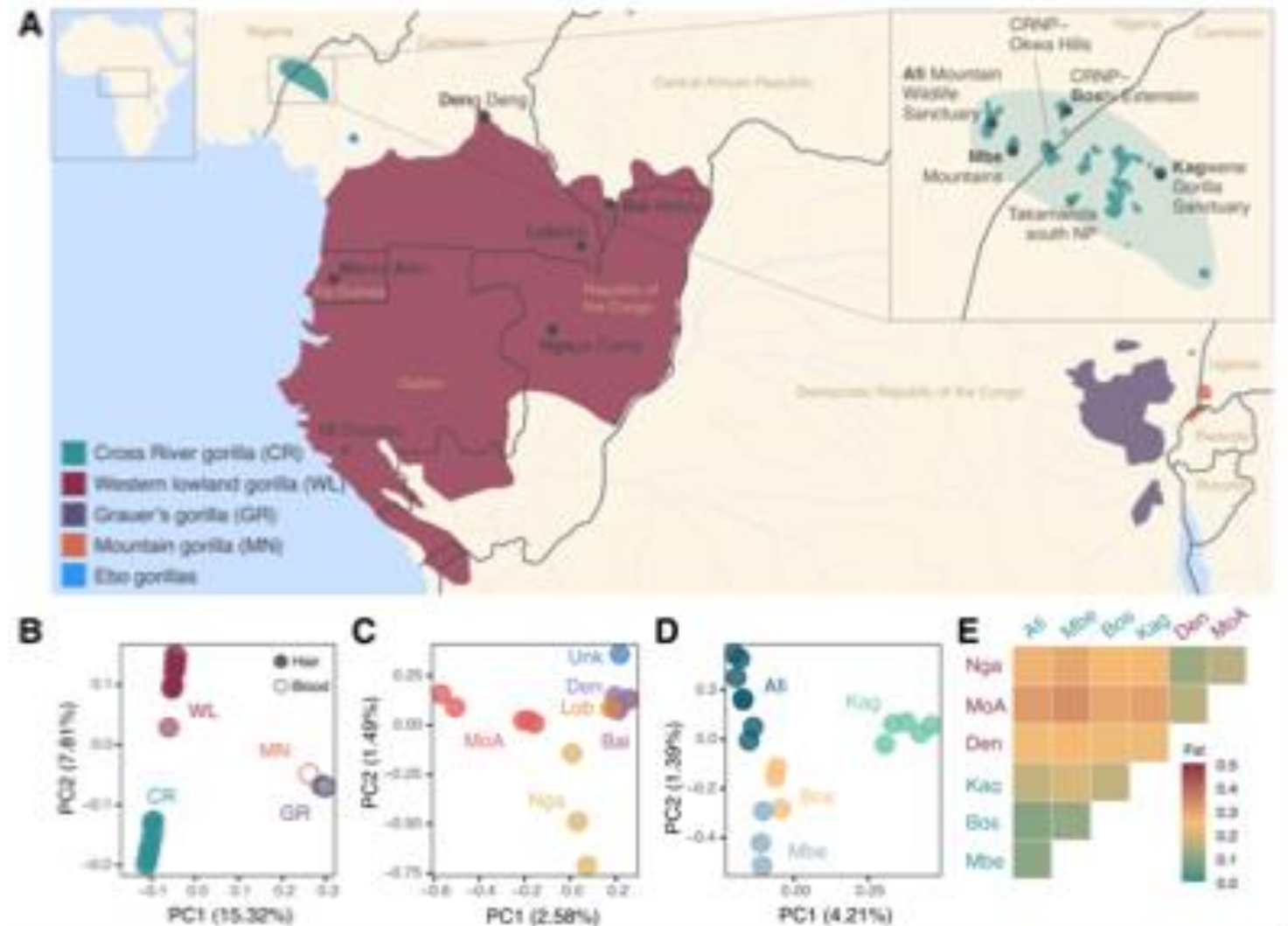
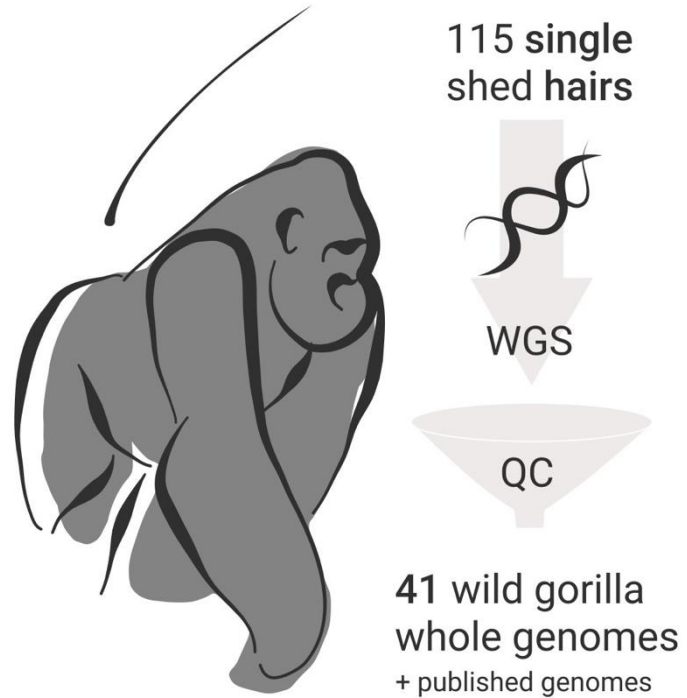
Social organization can limit access to mates and resources, affecting which alleles are passed on.

- ✓ Key idea: Social systems influence opportunities to reproduce.

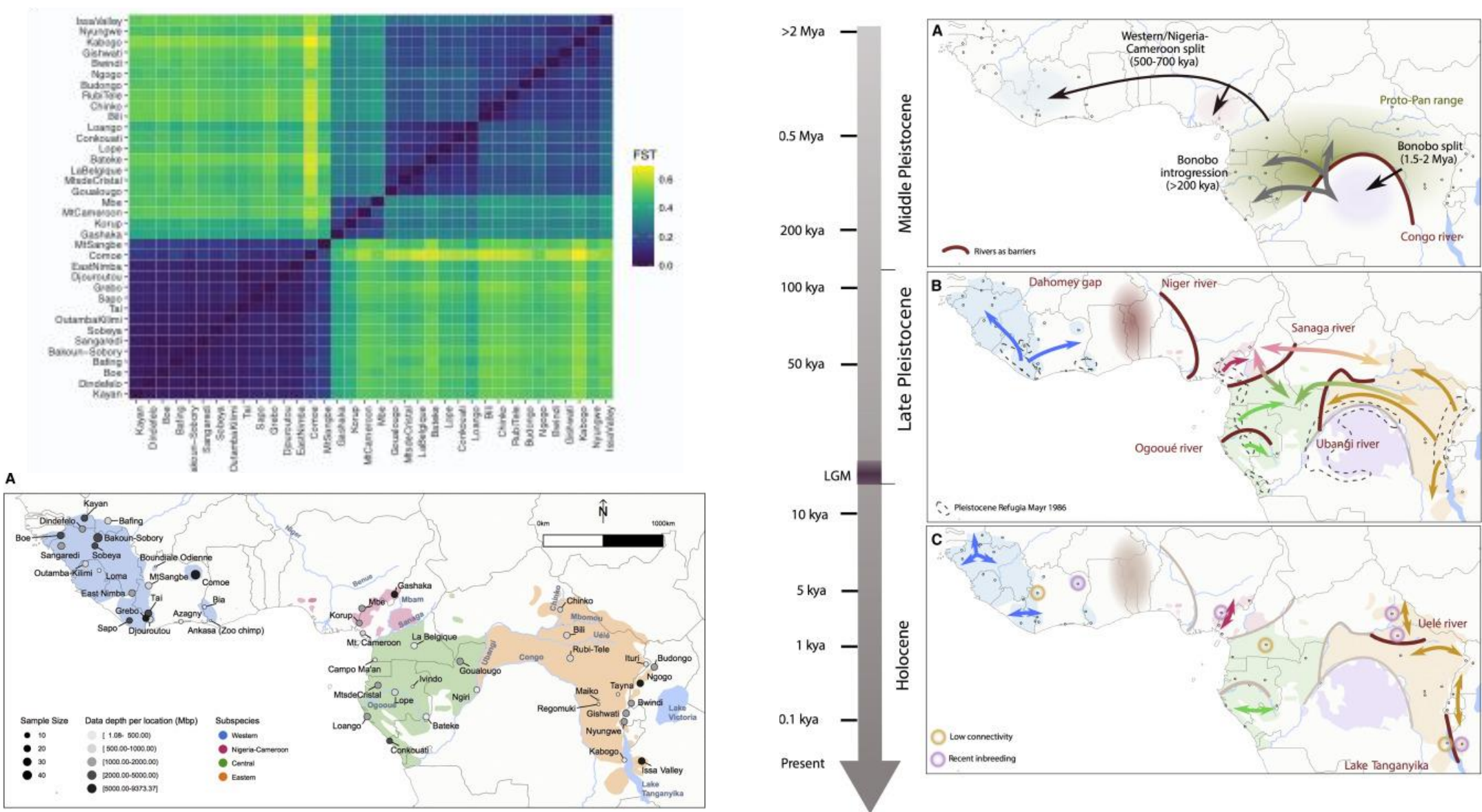
Examples – Saola - fragmentation



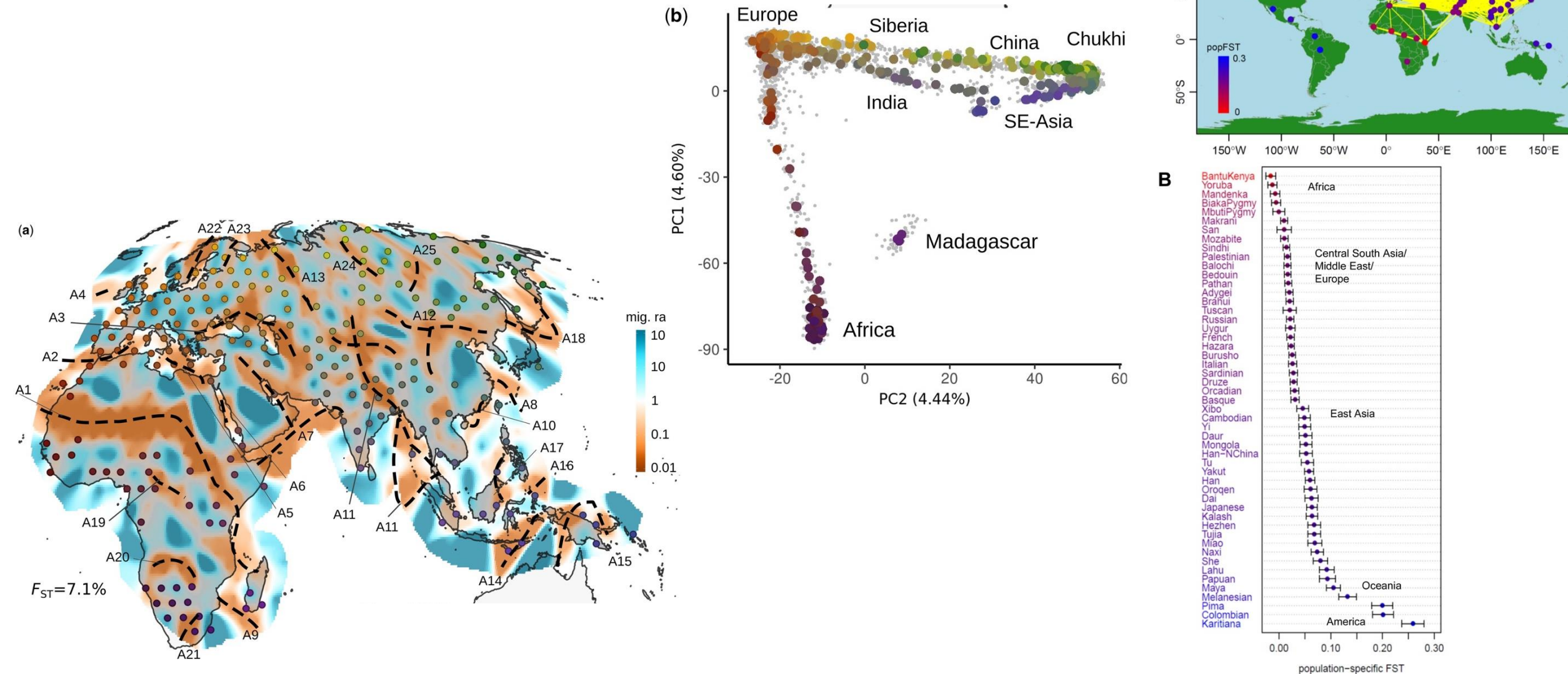
Examples – Gorillas - fragmentation



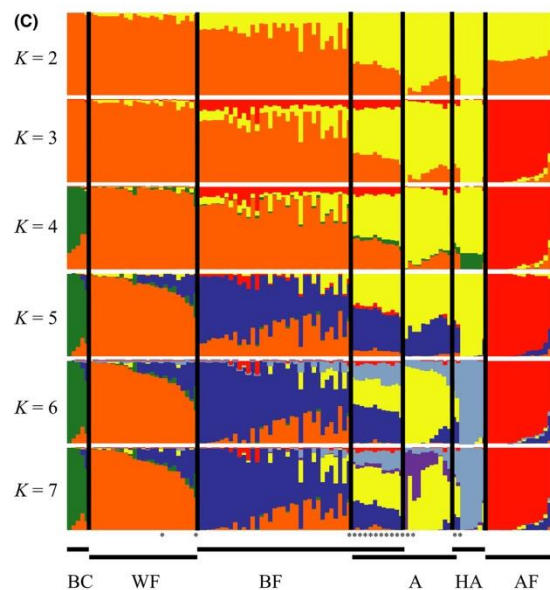
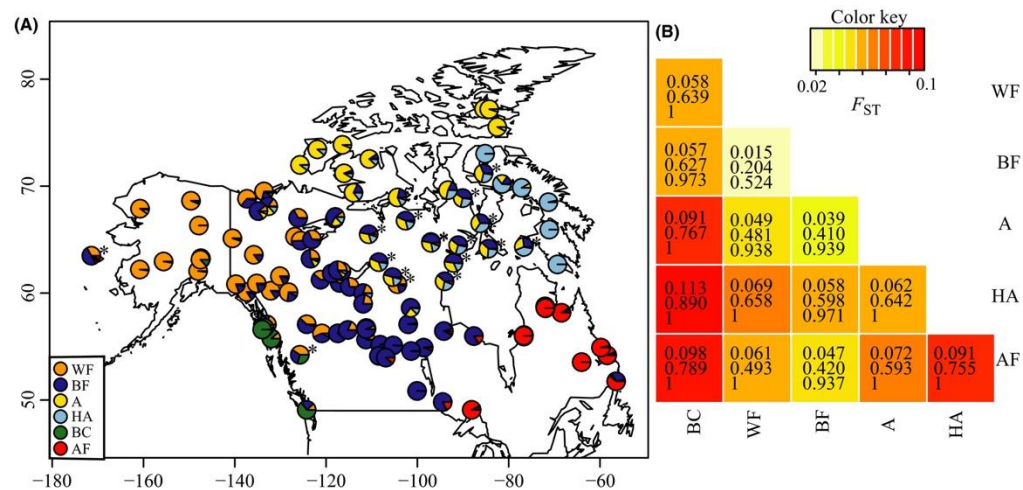
Examples – Chimps - barriers



Examples – Humans – barriers

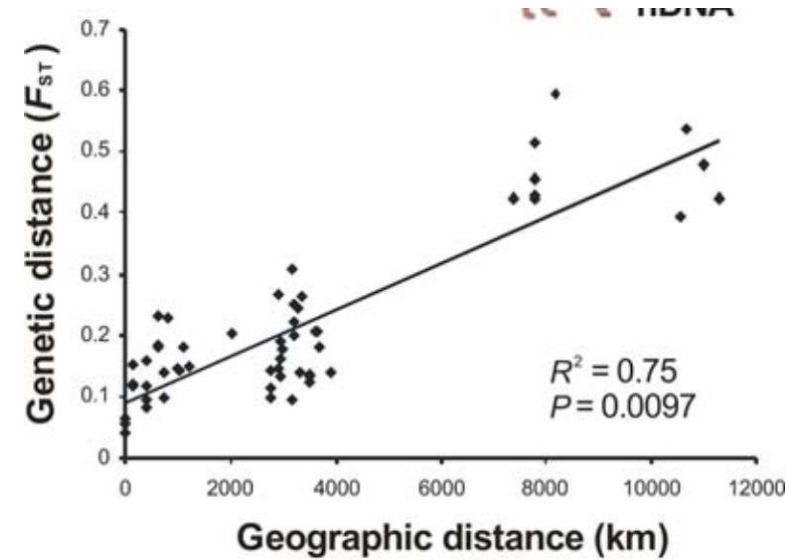
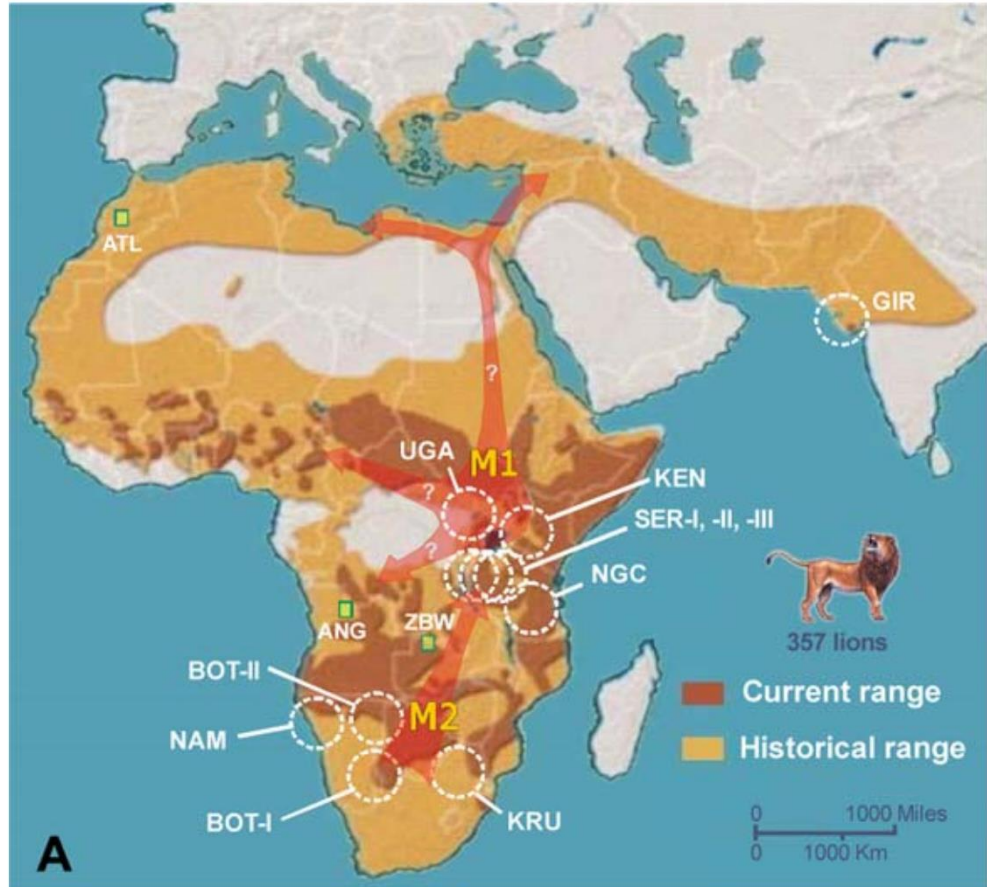


Examples – Wolves – local adaptation



West Forest	Cardiovascular system	Abnormality of the c
	Hearing	Functional abnormal
	Membranes	Integral component
	Metabolism	Metabolic pathways
	Organismal system	Abnormality of the liver
	Skeletal morphology	Abnormality of the external nose
	Vision	Abnormality of the eye
Boreal Forest	Immune response	Immune system processa
	Metabolism	Lipid metabolic processa
	Organismal system	Tissue developmenta
	Response to environment	Response to external stimulusa
	Skeletal morphology	Ossificationa
Arctic	Immune response	Positive regulation of lymphocyte mediated immunity
	Musculature	Abnormality of the musculature
	Organismal system	Functional abnormality of bladder
	Skeletal morphology	Abnormal bone ossification
High Arctic	Brain function	Learning or memory
	Metabolism	Histidine metabolism
British Columbia	Dentition	Misalignment of teeth
	Diet	Salivary secretion
	Metabolism	Arachidonic acid metabolism
	Musculature	Muscle hypertrophy
	Organismal system	Protein transporta
	Skeletal morphology	Disproportionate short stature
Atlantic Forest	Vision	Aplasia/hypoplasia of the iris
	Dentition	Hypodontia
	Metabolism	Glutathione metabolism
	Organismal system	Calcium ion transmembrane transporter activity
	Skeletal morphology	Aplasia involving forearm bones

Examples – Lions – Geographical distance



Scales of FST Estimation

Global

Single FST value across all populations. Useful for overall differentiation. Computed by averaging across loci with variance components. **Comparing species**

Example:
FST = 0.05 across 3 populations

Pairwise

FST between each pair of populations. Reveals which populations are most/least differentiated. **Comparing populations**

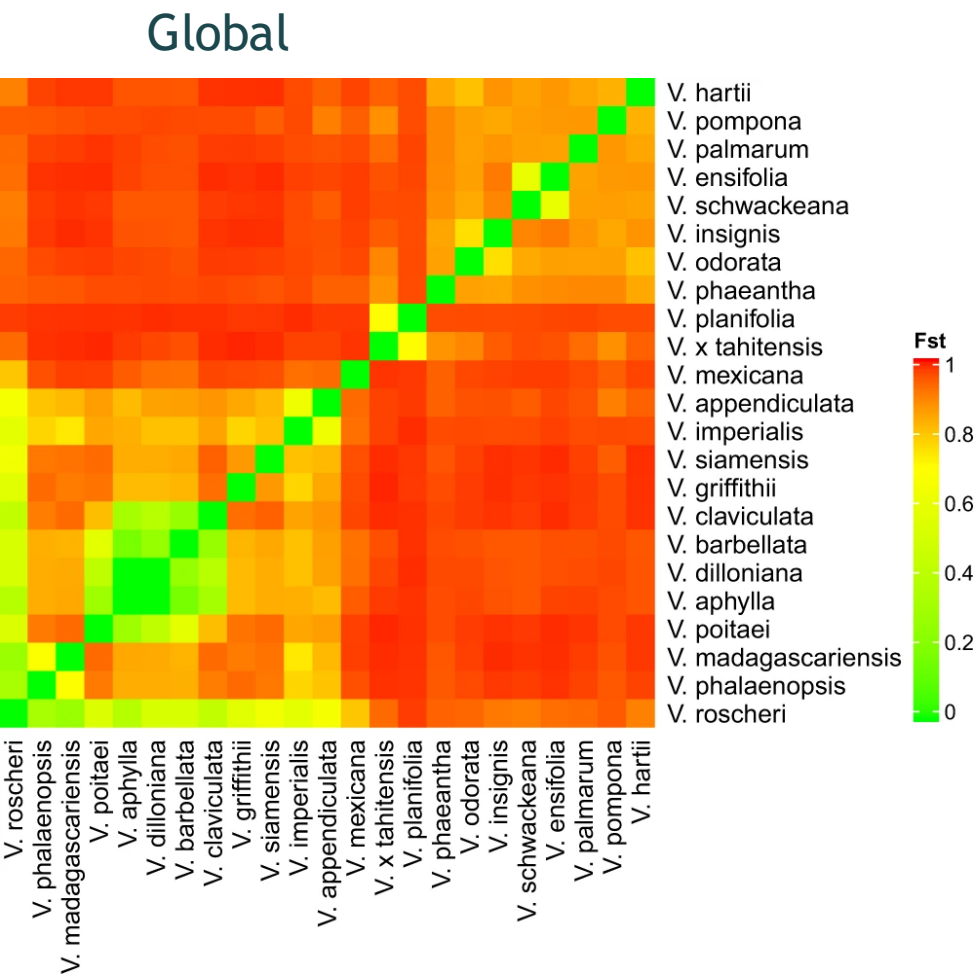
Example:
Pop1-Pop2: 0.03, Pop1-Pop3: 0.12

Per-Site / Windowed

FST at each variant or in sliding windows. Used for outlier scans to detect selection.

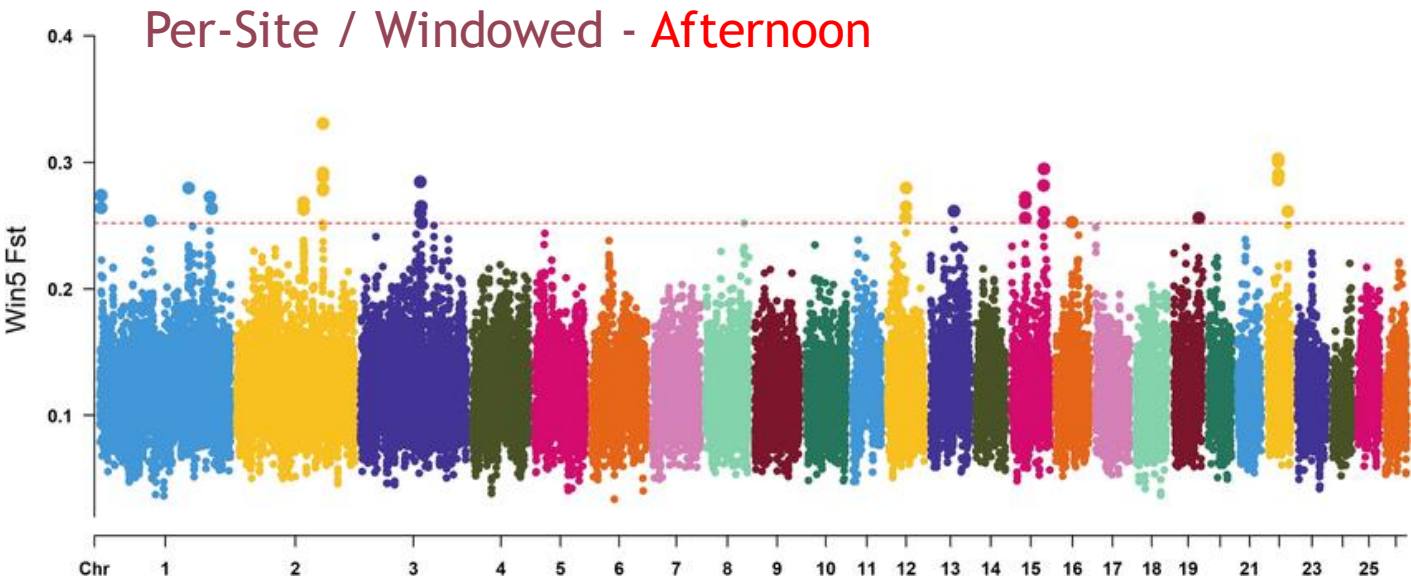
Example:
Window 1Mb: FST = 0.15 (potential outlier)

Scales of FST Estimation



Pairwise
Example of
Giraffes

West African	-	-	-	-	-	-	-	-	-
Kordofan	0.24	-	-	-	-	-	-	-	-
Nubian	0.24	0.16	-	-	-	-	-	-	-
Reticulated	0.35	0.33	0.32	-	-	-	-	-	-
Masai	0.64	0.63	0.62	0.55	-	-	-	-	-
Masai-Selous	0.67	0.66	0.65	0.58	0.10	-	-	-	-
Southern African	0.70	0.69	0.68	0.62	0.63	0.67	-	-	-
Southern Central	0.69	0.68	0.67	0.60	0.62	0.66	0.10	-	-
Angolan	0.71	0.71	0.70	0.63	0.66	0.69	0.19	0.14	-



FST Pitfalls – What can go wrong?

1 Sample size imbalance

Unequal N between populations biases F_{ST} upward for the smaller group.

- Use jackknifing or Weir & Cockerham estimator which partially corrects for this.

2 Ascertainment bias

SNPs discovered in one population are enriched for high-frequency derived alleles in that population, artificially inflating F_{ST} when comparing to others.

- Less relevant with whole-genome sequencing, but still present in some reference panels.

3 Heterozygosity dependency

Its maximum possible value at a locus depends on the frequency of the most common allele and on the number of alleles; for very common or very rare alleles, the attainable F_{ST} can be far below 1

- Low-diversity populations can show inflated F_{ST} even with minor allele frequency differences.

4 Admixture & structure within populations

Cryptic structure within a "population" inflates F_{ST} . Always check PCA first. Relatedness within populations also biases estimates.

- Run PCA first and clean your dataset of related individuals

Time for exercises

Fst in Wildebeest



Black Wildebeest



Blue Wildebeest